

NET731 FA3

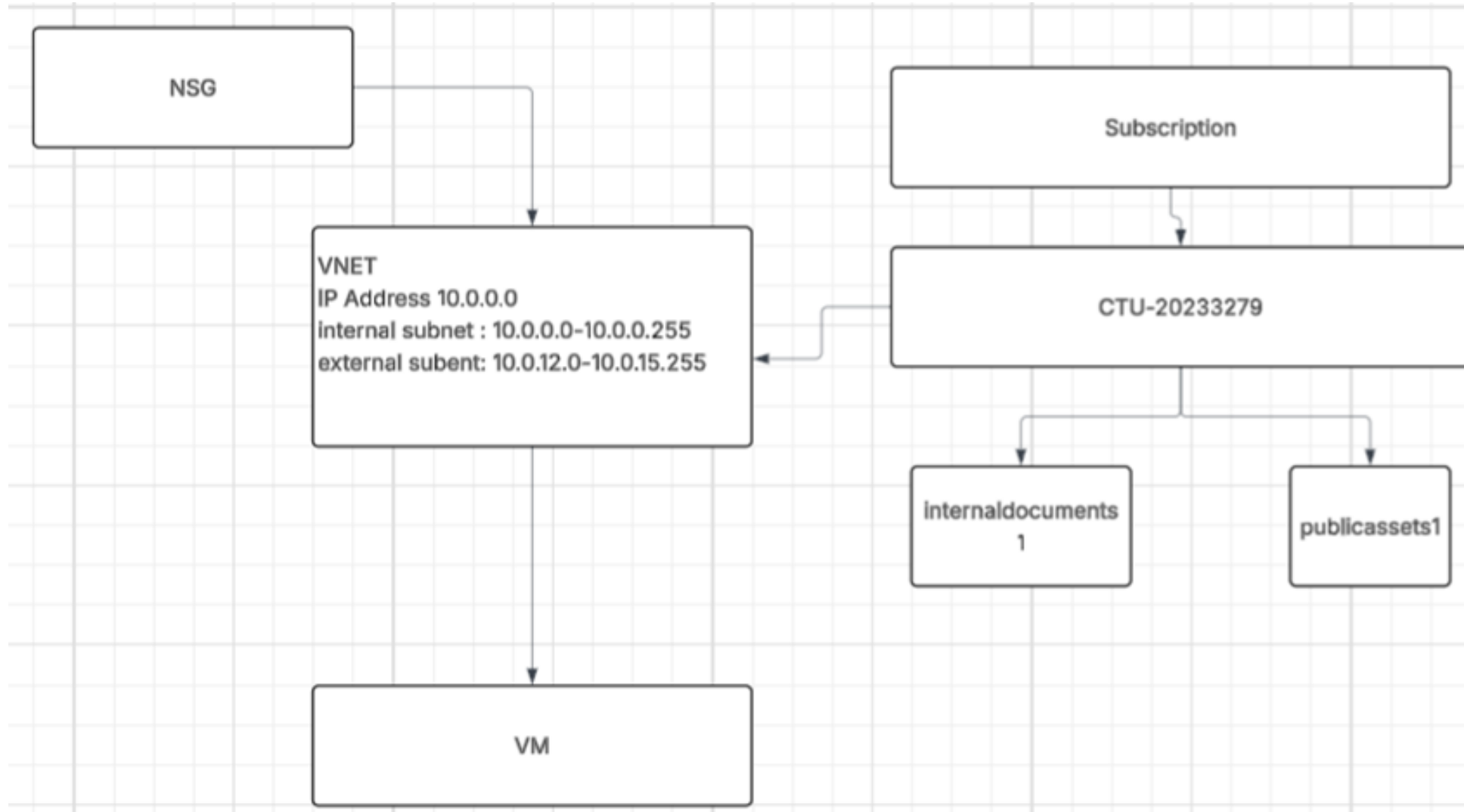
Heinrich Cornelius Meyer 20233279

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Project Overview

- I Built a very rudimentary azure environment containing a VM, VNet, NSG and a few users. The main purpose of the environment was to learn how to practically interact with the environment

Architecture diagram



Architecture diagram explained

- It's a simple diagram showing how the CTU-20233279 resource group is within a subscription, the two storage accounts associated with the resource group, a VNet and the respective NSG and VM associated with that VNet. It also shows the IP and subnet ranges for the VNet

Key technical decisions

- Not being able to create a student account for the azure environment(used the credit for last years modules), I improvised and switched over to a pay as you go account. Trying to work as cheaply as possible while still completing my practical.
- Another key choice was the design of my NSG's I tried to keep them basic. I was focusing more on how to stop the general threats you would encounter.

Security and monitoring

- I implemented a few inbound and outbound NSG rules to prevent basic threats
- I used RBAC to enforce the principles of least privilege to my users

Cost management

- Because of my account limitations I tried to complete my practical as cheaply as possible. I ended up not having to pay at all, due to how efficiently I managed my resource spending

Cost management

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Heinrich Meyer | Cost analysis

Billing account

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Save



Save as



Delete view



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Cost by resource



Configure billing account



Feedback



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Help

Overview

Access control (IAM)

Billing scopes

Diagnose and solve problems

Cost management

Cost analysis

Cost alerts

Budgets

Advisor recommendations

Billing

Invoices

Payment methods

Payment history

Reservation transactions

Billing profiles

Benefits

Products + services

ACTUAL COST (USD)

\$0.01

FORECAST UNAVAILABLE

--

BUDGET: NO_CASH

\$50 /yr

Group by: None

Granularity: Accumulated

Area

\$50

\$40

\$30

\$20

\$10

\$0

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

Dec

Accumulated cost

Annually budget

Overage

Service name



Virtual Machines

\$0.02

Virtual Network

<\$0.01

Storage

<\$0.01

No service name

-\$0.02

Location



za north

\$0.03

No resource location

-\$0.02

Subscription



Azure subscription 1

\$0.03

No subscription ID

-\$0.02

Challenges and solutions

- During week 7 we were required to collect metrics from our environment, however because I had ensured to stop (or delete where necessary) any resources that could generate costs I did not have any metrics to measure.
- I overcame this by creating new resources and running them for a very short amount of time so that there was at least something to monitor

Lessons learned

- I feel as if I have become more knowledgeable on the practical ways to use the azure portal. At the very least I feel comfortable with the basics.

Conclusion

- Azure is a powerful platform that if utilized properly can create a very efficient environment that could exceed your business needs. However, you should take time planning and creating your environment to ensure everything will function the way you want.
- Its also important to remember that most issues can be easily researched on google, don't mindlessly struggle when you could find the solution online